

WHITE SANDS, NM, USA

WMO: 722690

Lat: 32.383N

Lon: 106.483W

Elev: 1244

StdP: 87.25

Time Zone: -7.00 (NAM)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -7.5	(c) -5.3	(d) -16.8	(e) 1.0	(f) 6.0	(g) -14.7	(h) 1.2	(i) 6.4	(j) 10.4	(k) 12.8	(l) 8.5	(m) 12.4	(n) 0.8	(o) 180	(p) 0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	37.2	17.6	35.8	17.7	34.6	17.7	21.0	30.8	20.5	30.1	20.0	29.3	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	15.9	22.3	18.1	15.2	22.4	17.5	14.6	22.6	66.5	30.9	64.5	30.0	62.6	29.0	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.2	5.9	-12.0	39.5	3.1	1.4	-14.3	40.5	-16.1	41.3	-17.8	42.0	-20.1	43.0
			-12.9	22.8	2.8	0.9	-14.9	23.4	-16.5	23.9	-18.1	24.4	-20.2	25.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.5	8.7	12.5	17.0	21.5	25.7	26.8	25.8	22.6	17.0	10.1	5.5
	DBStd	8.37	3.62	3.84	3.82	4.10	3.20	2.76	1.89	2.05	2.91	3.31	4.22	4.12
	HDD10.0	429	144	62	19	5	0	0	0	0	0	2	52	146
	HDD18.3	1638	398	270	183	67	9	0	0	0	4	61	249	397
	CDD10.0	2839	4	26	97	216	357	472	521	489	377	221	53	7
	CDD18.3	1006	0	0	2	29	108	222	263	231	131	21	1	0
	CDH23.3	12796	0	11	95	522	1537	3125	3142	2467	1447	431	20	0
	CDH26.7	6263	0	1	12	139	658	1781	1706	1233	622	110	2	0
Wind	WSAvg	2.0	1.6	2.1	2.9	3.0	2.7	2.2	2.0	1.7	1.5	1.5	1.8	1.6
Precipitation	PrecAvg	266	12	12	6	9	13	19	53	49	38	24	11	18
	PrecMax	427	42	62	22	49	70	95	158	170	96	83	46	103
	PrecMin	129	0	0	0	1	0	1	16	22	1	0	0	0
	PrecStd	71	13	14	6	11	16	21	31	33	24	21	11	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	25.3	28.1	32.2	36.0	39.6	38.7	36.3	35.9	31.3	25.7	20.0
		MCWB	8.2	11.0	11.8	14.0	15.2	18.3	18.5	19.5	18.7	14.2	12.2	8.8
	2%	DB	17.8	21.6	25.9	29.9	33.8	37.9	36.9	35.0	33.6	29.5	23.3	17.7
		MCWB	7.7	9.6	11.1	13.0	15.0	17.3	18.0	19.4	17.8	14.4	11.3	8.3
	5%	DB	15.6	19.6	24.0	28.2	32.2	36.2	35.6	34.0	32.0	27.7	21.4	15.8
		MCWB	6.8	8.4	10.2	12.4	14.4	16.7	18.2	19.2	17.6	14.1	10.5	7.4
	10%	DB	13.7	17.7	22.0	26.5	30.5	34.5	34.2	32.8	30.4	25.9	19.3	13.9
		MCWB	6.1	7.8	9.4	11.7	14.0	16.5	18.4	19.3	17.2	13.8	9.5	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.7	13.1	15.5	18.8	20.8	21.9	21.9	20.8	17.9	14.8	11.7
		MCDB	15.0	21.7	24.9	28.4	29.4	32.4	31.5	31.4	30.1	23.9	19.4	16.1
	2%	WB	9.0	10.5	11.8	14.1	16.8	19.8	20.7	21.3	20.0	16.6	13.0	9.8
		MCDB	14.7	19.8	24.0	27.0	28.8	30.7	30.6	30.8	28.7	23.4	19.3	13.8
	5%	WB	7.9	9.4	10.8	13.1	15.8	19.0	20.2	20.7	19.3	15.5	11.8	8.7
		MCDB	13.8	17.8	22.8	25.3	27.8	30.1	30.1	30.2	27.5	23.5	19.2	13.9
	10%	WB	6.8	8.3	9.8	12.3	15.1	18.2	19.7	20.2	18.7	14.7	10.5	7.4
		MCDB	12.9	16.5	20.9	24.7	26.8	29.4	29.7	29.1	26.7	23.1	18.2	13.0
Mean Daily Temperature Range	5% DB	MDBR	14.9	15.6	16.3	16.5	16.3	16.9	14.7	13.7	14.8	16.0	15.5	14.3
		MCDBR	18.3	19.2	19.7	19.1	18.3	19.1	16.8	15.6	16.8	19.7	18.1	17.8
		MCWBR	10.7	10.2	9.7	8.6	7.7	7.4	5.9	4.8	6.1	8.5	9.2	10.4
	5% WB	MCDBR	15.6	16.6	18.2	17.9	16.5	15.7	14.5	13.4	13.5	15.4	15.1	14.2
		MCWBR	9.6	9.0	9.3	8.3	6.9	5.8	4.9	4.2	5.0	6.9	8.1	9.0
Clear-Sky Solar Irradiance	taub	0.244	0.250	0.267	0.296	0.315	0.333	0.369	0.348	0.322	0.278	0.258	0.247	
	taud	2.651	2.633	2.550	2.436	2.433	2.416	2.396	2.436	2.477	2.609	2.607	2.635	
	Ebn at Noon	1005	1026	1021	993	969	946	910	928	945	977	977	984	
	Edh at Noon	73	83	97	114	116	117	119	113	104	84	76	71	
All-Sky Solar Radiation	RadAvg	3.57	4.57	5.91	7.23	7.93	7.95	7.14	6.60	5.88	4.88	3.83	3.22	
	RadStd	0.24	0.34	0.31	0.29	0.26	0.33	0.34	0.24	0.26	0.36	0.23	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	N/A

Nomenclature: See separate page